Lipid-Lowering Therapy Use Among Cancer Survivors Undergoing Lung Cancer Screening: A Cross-Sectional Analysis

SITEMAN CANCER CENTER

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Background

- Cardiovascular disease (CVD) causes 56% of non-cancer deaths among cancer survivors, yet lipid-lowering therapy (LLT) remains underutilized.
- Lung cancer screening (LCS) populations have dual high-risk profile: heavy smoking exposure and cancer history.
- National data show profound disparities: Black patients 48% less likely to be offered statins, rural residents 28% lower use, uninsured 3-4 fold lower odds.

Methods

- Cross-sectional study of 7,778 cancer survivors (50–80 years) who underwent lung cancer screening in Barnes- Jewish Healthcare's LCS program between 2015-2023.
- Primary outcome was LLT use (statins or statin-substitute) documented in electronic health records.
- Associated factors included demographics (age, sex, race), clinical factors (prior ASCVD, cancer type), healthcare access (insurance, rurality via RUCA) and Area Deprivation Index.
- Analysis: Multivariable logistic regression, adjusted for demographics, insurance, smoking, indication, provider; stratified by race & sex.

Results: LLT Prevalence and associated factors

- Compared with lung cancer, **prostate** (AOR = 1.35; 95% CI 1.08–1.71), **hematologic** (AOR = 1.35; 95% CI 1.03–1.76), and **other solid** cancers (AOR = 1.23; 95% CI 1.07–1.42) were **associated with higher LLT use**.
- **Race-stratified: Among White patients only**, **male sex** was associated with higher LLT use (AOR = 1.13; p = 0.036) and **rural residence** with lower LLT use (AOR = 0.81; p = 0.005); these were **not significant among Black patients**.
- Prior ASCVD was the strongest associated factor (AOR=2.54, p<0.001), followed by age ≥65 years (AOR=1.41, p<0.001).
- Contrary to robust national disparities, race showed NO association with LLT use (p=0.601).
- Rural residence associated with 29% lower odds of LLT use (AOR=0.71, p<0.001) even after adjusting for all other factors.
- Insurance status (p=0.313) and neighborhood deprivation (ADI, p=0.292) showed NO association with LLT use.

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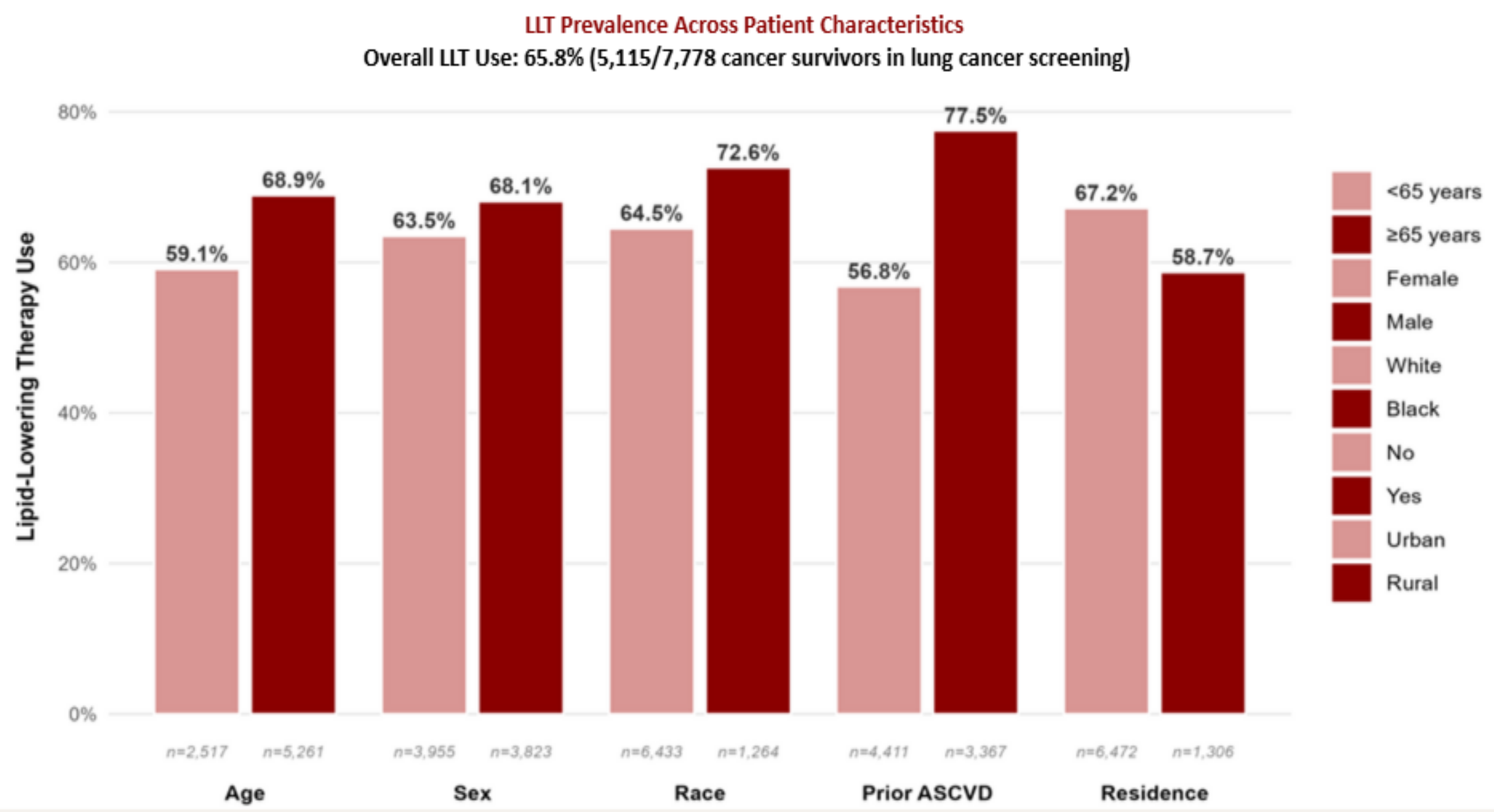
Results

Table 1: Demographic characteristics (N=7,778.)

Variable	Category	Count (%)
Age	Above 65	5,261 (68%)
	Below 65	2,517 (32%)
Sex	Female	3,955 (51%)
	Male	3,823 (49%)
Insurance	Medicaid	8 (0.1%)
	Medicare	3,993 (51%)
	Private	3,785 (49%)
	Multiple	796 (10%)
Race	Black	1,264 (16%)
	White	6,433 (89%)
	Asian	44 (1%)
	Other	37 (<1%)
Prior ASCVD	Yes	3,367 (43%)
	No	4,411 (57%)
Cancer Type	Lung	933 (17%)
	Breast	1,303 (10%)
	Prostate	575 (7%)
	Hematologic	343 (4%)
	Colorectal	216 (3%)
	Other solid	3,148 (40%)
	Multiple	1,398 (18%)
Rural/Urban	Rural	1,306 (17%)
	Urban	6,472 (83%)
LLT status	On LLT	5,115 (66%)
	Not on LLT	2663 (34%)
ADI	Q1 (Least)	1,442(19%)
	Q2	1,493 (19%)
	Q3	1,579 (20%)
	Q4	1,569 (20%)
	Q5 (Most)	1,695(22%)

Table 1 abbreviations: ADI = Area Deprivation Index, LLT = Lipid Lowering Therapy, ASCVD = Atherosclerotic Cardiovascular Disease, RUCA = Rural-Urban Commuting Area.



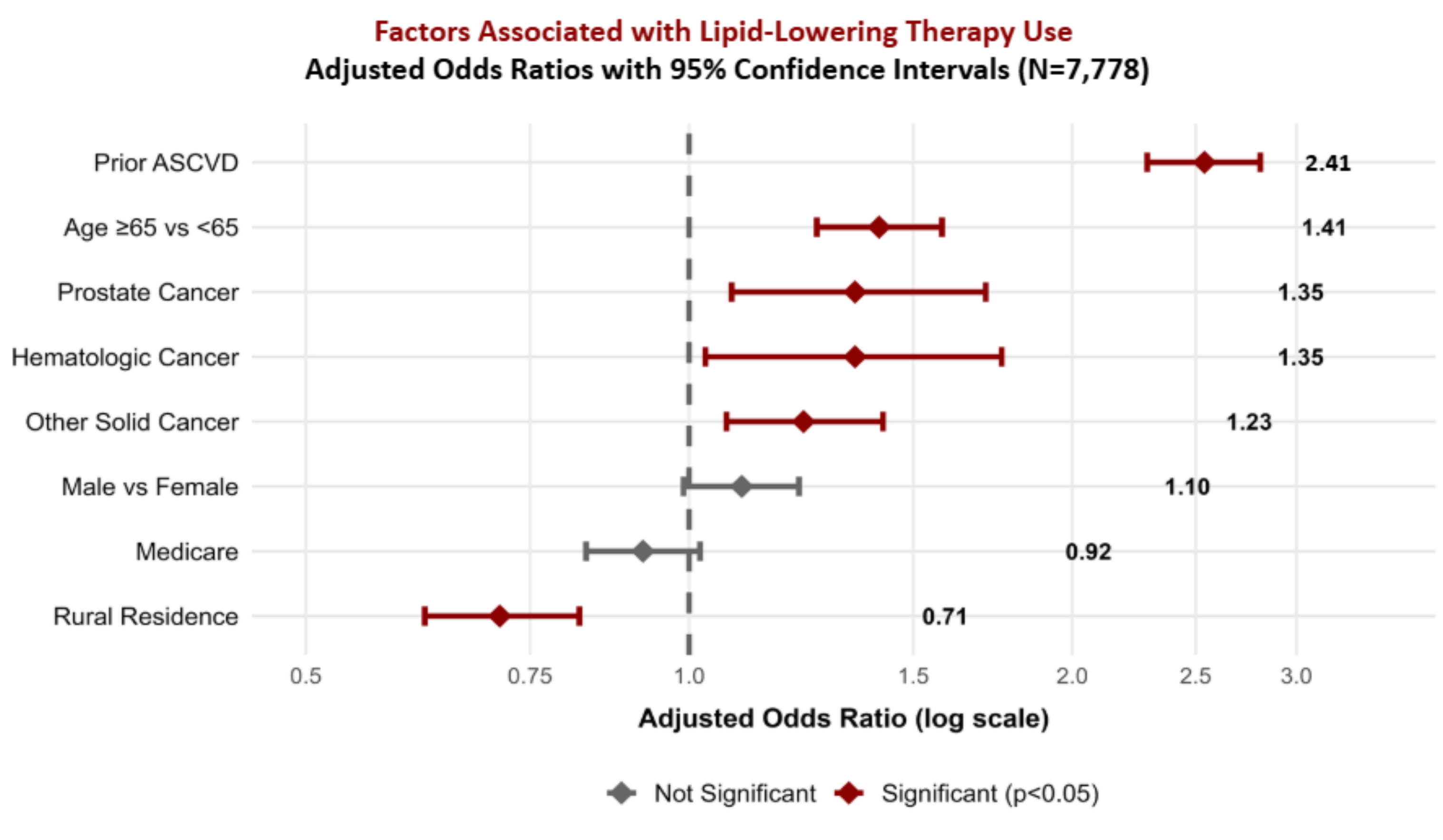


Table 2: Multivariable Associated factors of LLT Use Among Cancer Survivors in Lung Cancer Screening

Variable	OR	95% CI	p-value
Demographics			
Age ≥65 vs <65	1.41	(1.26 - 1.58)	<0.001
Male vs Female	1.10	(0.99, 1.22)	0.082
Race (Overall)	---	---	0.601
Clinical Factors			
Prior ASCVD (Yes vs No)	2.54	(2.29, 2.81)	<0.001
Cancer: Prostate vs Lung	1.35	(1.03, 1.76)	0.013
Cancer: Hematologic vs Lung	1.35	(1.08-1.71)	0.009
Cancer: Other solid vs Lung	1.23	(1.07-1.42)	0.004
Social Determinants			
Rural Vs Urban	0.71	(0.62-0.82)	<0.001
Insurance (overall)	---	---	0.313
ADI Quintile (overall)	---	---	0.292

Abbreviations: ADI = Area Deprivation Index, CI = Confidence Interval,

Table 3: Race-Stratified Analysis of Key LLT Predictors

Variable	Black (n=1,264)		White (n=6,433)	
	OR	P	OR	P
Age ≥65	**1.51**	**0.004**	**1.39**	**<0.001**
Male sex	1.0	>0.9	**1.13**	**0.036**
Prior ASCVD	**2.64**	**<0.001**	**2.51**	**<0.001**
Rural	0.54	0.3	**0.81**	**0.005**

Discussion

- **Lipid-lowering therapy remains underused** among cancer survivors at high cardiovascular risk, even during routine lung cancer screening.
- **Rural patients were less likely** to receive therapy, underscoring persistent geographic gaps in preventive care.
- **Prior ASCVD strongly predicted use**, suggesting LLT is still primarily used reactively rather than preventively.
- **Improving follow-up and care coordination** during screening could help close prevention gaps and reduce heart disease deaths in cancer survivors.